MENG WANG

Department of Civil, Architectural and Environmental Engineering
The University of Texas at Austin
Phone: (+1) 832-441-5726 | Email: meng.wang@austin.utexas.edu
Google Scholar Profile | Personal Website

RESEARCH INTERESTS

My research centers on molecular interactions and electrolyte thermodynamics in aqueous systems, with a focus on advancing water and energy sustainability. I investigate how interactions among ions, water, polymer coatings, and mineral surfaces govern speciation, sorption, and transport, ultimately controlling chemical reactions in aqueous and interfacial environments. By integrating spectroscopy and x-ray scattering, thermodynamic and computational modeling, and multiscale experiments, I connect molecular-scale interactions to macroscopic processes governing membrane-based separations, critical mineral recovery, and fouling/scaling control across water–energy systems.

Research keywords: ion association and solvation; interfacial chemistry; ion transport; polymer-functionalized membrane interfaces; predictive thermodynamic modeling; sustainable separations and resource recovery.

EDUCATION

University of Houston ***Houston, TX***
Ph.D., Environmental Engineering 2023

- Dissertation: "Mechanistic Study on Polymer Coatings to Control Mineral Scaling in Desalination Treatment"

Georgia Institute of Technology ***Atlanta, GA***
M.S., Computer Science 2026

China University of Petroleum, Beijing ***Beijing, China***
M.S., Petroleum Engineering 2018

- Thesis: "Numerical Study of Proppant Transport with Supercritical CO_2 in Fracture"

Northeast Petroleum University ***Heilongjiang, China***
B.E., Petroleum Engineering 2015

- Thesis: "Hydraulic Fracturing Model Featuring Initiation Beyond the Wellbore Wall for Directional Well in Coal Bed"

HONORS & AWARDS

8. **Early Career Postdoctoral Fellows Program**, The University of Texas at Austin, 2026
7. **American Chemical Society GEOC Early Career Award**, American Chemical Society, 2025
6. ***Processes* Outstanding Reviewer Award**, MDPI, 2025
5. **Graduate Student Travel Award**, ACS Southwest Regional Meeting, 2021
4. **Graduate Tuition Fellowship,** University of Houston**,** 2018–2023
3. **Robert Rocca Scholarship**, Tenaris China, 2017
2. **First-Class Scholarship**, China University of Petroleum, Beijing, 2017
1. **Outstanding Graduate Student**, China University of Petroleum, Beijing, 2016

RESEARCH EXPERIENCE

The University of Texas at Austin *08/2023 – Present*

Postdoctoral Researcher | Advisor: Dr. Lynn E. Katz

- Developed a molecular transport framework linking ion association, hydration, and thermodynamic nonideality to aqueous speciation, sorption, and transport
- Established complementary Raman and dielectric spectroscopy methods to quantify ion pairing and ion–solvent dynamics in sulfate electrolytes containing mono-, di-, and trivalent cations (e.g., Li^+, Mg^{2+}, and La^{3+})
- Developed a computational pipeline (dielectric analyzer) for dielectric-spectrum fitting and relaxation-process deconvolution
- Resolved discrepancies in sulfate ion-pair stability constants across measurement methods and applied the validated constants to real-world water matrices, quantifying the effect of ion pairing on sulfate speciation
- Developed a spectroscopically constrained extended Pitzer model that incorporates ion-pairing equilibria, predicts activity coefficients of charged ion pairs, and reveals systematic bias in conventional models that neglect ion pairing
- Contributed to a pilot-scale produced-water separation project by evaluating ceramic and hollow-fiber membrane performance through solution-chemistry analysis and surface characterization
- Authored **3** first-author manuscripts (**1** under review, and **2** in preparation) and delivered **3** conference presentations

University of Houston *08/2018 – 05/2023*

Graduate Research Assistant | Advisor: Dr. Debora F. Rodrigues

- Developed a mechanistic framework linking polymer chemistry and interfacial hydration to resistance to organic fouling, biofouling, and scaling (e.g., gypsum) in reverse-osmosis (RO) systems
- Performed cross-flow RO experiments on polymer-coated membranes, characterizing fouling behavior and mineral scaling at organic–water interfaces to evaluate membrane antifouling and anti-scaling performance
- Integrated membrane-performance measurements with surface and scattering characterization to establish structure–scaling relationships for antifouling and anti-scaling interface design
- Published **4** first-author (**1** in preparation), and **3** co-authored papers and presented findings at **2** professional meetings

Argonne National Laboratory—Advanced Photon Source (APS) *01/2019 – 11/2022*

Visiting Student (concurrent with Ph.D. research at University of Houston)

- Served as the primary writer for two General User Proposals that secured 36 total shifts of beamtime at APS
- Executed *in situ* SAXS/GISAXS and *ex situ* GIWAXS experiments at 12-ID-B to quantify early-stage mineral nucleation, growth, and crystallinity on polymer-coated interfaces
- Processed and modeled X-ray scattering datasets using MATLAB, Igor Pro, and Irena SAS; the results contributed to a publication in *Water Research*

China University of Petroleum, Beijing *09/2015 – 07/2018*

Graduate Research Assistant | Advisor: Dr. Haizhu Wang

- Developed computational fluid dynamics (CFD) models of supercritical CO_2–proppant transport in fractures and used them to identify design parameters governing particle placement in waterless fracturing
- Co-developed experimental and drilling devices, resulting in two granted Chinese patents
- Published **1** first-author and **6** co-authored papers on supercritical CO_2 transport, fracturing, and subsurface applications

Northeast Petroleum University *02/2015 – 07/2015*

Undergraduate Research Student | Advisor: Dr. Yuwei Li

- Developed a coal-bed fracture-initiation model and wrote a C/C++ simulator to predict fracture-initiation pressure along coal cleats near wellbores during hydraulic fracturing
- Published the model in the *Journal of Geophysics and Engineering*

PUBLICATIONS

Manuscripts Under Review and in Preparation:

4. **M. Wang**, Z. Huang, J. Cui, N. Vinh, B. Freeman, and L. Katz. Spectroscopic Insights into Ion Pairing in Sulfate Electrolytes: Implications for Sorption in Aqueous Systems. *Environmental Science & Technology* (2026). **[Accepted]**
3. **M. Wang**, L. Katz. Incorporating Ion Pairing into an Extended Pitzer Model for Activity Coefficients. *Chemical Engineering Journal*. **[In preparation]**
2. **M. Wang**, J. Cui, N. Vinh, and L. Katz. Unraveling Ion Pairing and Hydration Structure of Sulfate Salts in Aqueous Solution: From Monovalent to Trivalent Ions. *Environmental Science & Technology*. **[In preparation]**
1. **M. Wang**, R. Jacovone, L. Faria, X. Zuo, R. O'Hara, A. Asatekin, and D. Rodrigues. Zwitterion Loading Tunes Heterogeneous Gypsum Nucleation on Amphiphilic Copolymer Coatings. *Water Research*. **[In preparation]**

Peer-Reviewed Journal Articles:

21. R. Jacovone, **M. Wang**, J. Soares, S. Sakata, and D. Rodrigues. Investigating the Influence of Polymaleic Acid on Reverse Osmosis Membrane Performance to Mitigate Gypsum Scaling. *Desalination* 619 (2026): 119544. https://doi.org/10.1016/j.desal.2025.119544.
20. **M. Wang**, M. Sarma, S. Lounder, A. Mondal, L. Muthusamy, G. Koley, A. Asatekin, and D. Rodrigues. Organic Fouling on Zwitterionic Amphiphilic Copolymers: Implications in Biofouling. *ACS Applied Materials & Interfaces* 17(20) (2025): 30149–30160. https://doi.org/10.1021/acsami.5c07057.
19. **M. Wang**, X. Zuo, R. Jacovone, R. O'Hara, A. Mondal, A. Asatekin, and D. Rodrigues. Influence of Zwitterionic Amphiphilic Copolymers on Heterogeneous Gypsum Formation: A Promising Approach for Scaling Resistance. *Water Research* 266 (2024): 122439. https://doi.org/10.1016/j.watres.2024.122439.
18. **M. Wang**, H. Nguyen, S. Lounder, A. Asatekin, and D. Rodrigues. Calcium Sulfate Formation on Different Zwitterionic Amphiphilic Copolymer Substrates for Salt Water Treatment. *ACS Applied Polymer Materials* 4(10) (2022): 7090–7101. https://doi.org/10.1021/acsapm.2c01014.
17. H. Arabaghian, **M. Wang**, J. Ordonez, and D. Rodrigues. Zwitterionic Polymers in Biofouling and Inorganic Fouling Mechanisms. In *60 Years of the Loeb-Sourirajan Membrane* (2022): 33–70. https://doi.org/10.1016/B978-0-323-89977-2.00013-0. **[Book chapter; co-first author]**
16. **M. Wang**, B. Cao, Y. Hu, and D. Rodrigues. Mineral Scaling on Reverse Osmosis Membranes: Role of Mass, Orientation, and Crystallinity on Permeability. *Environmental Science & Technology* 55(23) (2021): 16110–16119. https://doi.org/10.1021/acs.est.1c04143.
15. A. Ansari, J. Peña-Bahamonde, **M. Wang**, D. Shaffer, Y. Hu, and D. Rodrigues. Polyacrylic Acid-Brushes Tethered to Graphene Oxide Membrane Coating for Scaling and Biofouling Mitigation on Reverse Osmosis Membranes. *Journal of Membrane Science* 630 (2021): 119308. https://doi.org/10.1016/j.memsci.2021.119308.
14. F. Chen, Y. Wang, R. Bi, Y. Pan, and **M. Wang**. Molecular Insights into CO_2-Enhanced Hydrocarbon Recovery and Its Sequestration in Multiscale Shale Reservoirs. *Chemical Engineering Journal* 498 (2024): 155913. https://doi.org/10.1016/j.cej.2024.155913.
13. Y. Wang, Q. Sun, F. Chen, and **M. Wang**. Multiscale Model for Hydrogen Transport and Storage in Shale Reservoirs. *SPE Journal* 29(6) (2024): 3238–3264. https://doi.org/10.2118/219472-PA.

12. T. Ge, **M. Wang**, F. Li, F. Min, and M. Zhang. Study and Spectroscopy Characterization of Carbon and Oxygen Structures of Gaoyang Coking Coal. *Journal of China Coal Society* 46(3) (2021): 1024–1031. https://doi.org/10.13225/j.cnki.jccs.2019.1693.
11. H. Wang, G. Li, Z. Shen, Z. He, Q. Liu, B. Zhu, Y. Wang, and **M. Wang**. Expulsive Force in the Development of CO_2 Sequestration: Application of SC-CO_2 Jet in Oil and Gas Extraction. *Frontiers in Energy* 13(1) (2019): 1–8. https://doi.org/10.1007/s11708-017-0458-6.
10. **M. Wang**, H. Wang, G. Li, B. Yang, Y. Zheng, and Q. Lu. Numerical Study of Proppant Transport with Supercritical CO_2 in Fracture. *China Petroleum Machinery* 46(11) (2018): 72–78, 84. https://doi.org/10.16082/j.cnki.issn.1001-4578.2018.11.012.
9. H. Wang, **M. Wang**, B. Yang, Q. Lu, Y. Zheng, and H. Zhao. Numerical Study of SC-CO_2 and Proppant Transport in Different Geometrical Fractures. *Greenhouse Gases: Science and Technology* 8(5) (2018): 898–910. https://doi.org/10.1002/ghg.1803.
8. H. Wang, G. Li, Z. He, S. Tian, **M. Wang**, B. Yang, Q. Lu, and L. Weng. Experimental Investigation on Abrasive Supercritical CO_2 Jet Perforation. *Journal of CO_2 Utilization* 28 (2018): 59–65. https://doi.org/10.1016/j.jcou.2018.09.018.
7. H. Wang, G. Li, Z. He, Z. Shen, X. Li, Z. Zhang, **M. Wang**, B. Yang, Y. Zheng, and L. Shi. Analysis of Rock Fracture Mechanism Caused by Supercritical CO_2. *Rock and Soil Mechanics* 39(10) (2018): 3589–3596. https://doi.org/10.16285/j.rsm.2017.0071.
6. H. Wang, G. Li, L. Zhao, Y. Wang, Y. Liu, Q. Liu, **M. Wang**, F. Gao, and M. Cai. Throttle Characteristics of Multi-Stage Circumfluence Nozzle During the Separate-Layer Injection of CO_2. *Petroleum* 4(2) (2018): 187–197. https://doi.org/10.1016/j.petlm.2017.08.003.
5. Y. Wang, J. Yuan, Z. Wang, Y. Yin, **M. Wang**, and W. Wang. Experimental Study on the Effect of Mechanical Vibration on Cement Stone Strength. *Natural Gas Exploration and Development* 41(1) (2018): 96–100.
4. H. Wang, G. Li, Z. He, Z. Shen, **M. Wang**, and Y. Wang. Mechanism Study on Rock Breaking with Supercritical Carbon Dioxide Jet. *Atomization and Sprays* 27(5) (2017): 383–394. https://doi.org/10.1615/AtomizSpr.2017017174.
3. X. Li, G. Li, H. Wang, S. Tian, X. Song, P. Lu, and **M. Wang**. A Unified Model for Wellbore Flow and Heat Transfer in Pure CO_2 Injection for Geological Sequestration, EOR, and Fracturing Operations. *International Journal of Greenhouse Gas Control* 57 (2017): 102–115. https://doi.org/10.1016/j.ijggc.2016.11.030.
2. Y. Li, D. Jia, **M. Wang**, J. Liu, C. Fu, X. Yang, and C. Ai. Hydraulic Fracturing Model Featuring Initiation Beyond the Wellbore Wall for Directional Well in Coal Bed. *Journal of Geophysics and Engineering* 13 (2016): 536–548. https://doi.org/10.1088/1742-2132/13/4/536.
1. X. Li, G. Li, H. Wang, P. Lu, S. Tian, R. Yang, and **M. Wang**. A Coupled Model for Predicting Flowing Temperature and Pressure Distribution in Drilling Ultra-Short Radius Radial Wells. *IADC/SPE Asia Pacific Drilling Technology Conference* (2016): SPE-180597-MS. https://doi.org/10.2118/180597-MS. **[Conference proceeding]**

FUNDING AND PROPOSAL EXPERIENCE

U.S. Department of Energy, Basic Energy Sciences (BES) — Energy Frontier Research Center *2026*

Center for Materials for Extraction and Energy Transformations (MXET)

- **Role:** Contributed to the development of the first research thrust through preliminary data analysis, figure preparation, experimental methodology development, and literature review to support the scientific rationale

  Status: Under review

  PI: Lynn E. Katz

Advanced Photon Source, Argonne National Laboratory — General User Proposal *2021*

Effect of Zwitterionic Copolymer Surfaces with Structural Variability on Heterogeneous Nucleation and Growth of $CaSO_4$ (GUP No. 77339)

- **Role:** Primary proposal writer. Independently developed and wrote the proposal and experimental plan

  Status: 27 shifts of beamtime at APS 12-ID-B over three visits

  PI: Debora F. Rodrigues

Advanced Photon Source, Argonne National Laboratory — General User Proposal *2020*

Calcium Sulfate Scaling on the Zwitterionic Amphiphilic Copolymer Substrates: Implication for Developing Anti-Scaling Membrane Surfaces (GUP No. 72353)

- **Role:** Primary proposal writer. Independently developed and wrote the proposal and experimental plan

  Status: 9 shifts of beamtime at APS 12-ID-B for remote-access experiments

  PI: Debora F. Rodrigues

National Natural Science Foundation of China (NSFC) — General Program *2018*

Mechanism of Proppant Transport in Fractures by Supercritical CO_2 (Grant No. 51874318)

- **Role:** Proposal contributor. Contributed substantially to proposal writing, including literature review, identification of research gaps, formulation of research objectives, and experimental design

  Status: Funded

  PI: Haizhu Wang

TEACHING EXPERIENCE

Co-instructor, Foundations of Environmental Engineering (**EVE 302**)
The University of Texas at Austin *01/2024–05/2024*

- Co-designed and delivered laboratory activities connecting course concepts with experimental practice
- Coordinated course examinations and supported consistent assessment of student learning
- Provided direct laboratory instruction and guidance on experimental procedures and course concepts

Teaching Assistant, Environmental Engineering (**CIVE 3331**)
University of Houston *08/2022–12/2022*

- Delivered problem-solving lectures explaining homework methods and course concepts
- Provided individualized support for students' conceptual and quantitative questions
- Evaluated homework and examinations with the instructor to maintain consistent grading

MENTORING EXPERIENCE

Undergraduate Students

- Denali Ibbotson, Colorado School of Mines, Summer 2019 (REU Program)
  Current position: Ph.D. student, Texas A&M University
- Satvik Raghunathan, The University of Texas at Austin, Summer 2024
  Current position: Undergraduate student, The University of Texas at Austin

Graduate Students

- Nikhil Khatri, University of Houston, Spring 2019
  Current position: Environmental Engineer, Distinct Engineering Solutions, Inc.
- Vicky Agrawal, University of Houston, Fall 2019
  Current position: Associate Consultant, WSP

- Raynara Maria Silva Jacovone, University of Houston, Fall 2021
  Current position: Postdoctoral Researcher, Clemson University
- Murchana Sarma, University of Houston, Fall 2022
  Current position: Ph.D. student, Clemson University
- Julia L. McGuire, The University of Texas at Austin, Fall 2023
  Current organization: Chicago Consulting Company

GRANTED PATENTS

2. H. Wang, G. Li, J. Li, Z. Huang, **M. Wang**, B. Yang, Q. Lu. A wire-driven drilling device. Patent No. CN106703720B, China University of Petroleum, Beijing, 2018.

1. H. Wang, G. Li, **M. Wang**, S. Tian, Z. Huang, B. Yang, Q. Lu, L. Weng. Supercritical CO_2sand-carrying simulation experimental device and system in fractures. Patent No. CN108590620B, China University of Petroleum, Beijing, 2020.

PRESENTATIONS

Oral Presentations(* indicates the presenter):

5. **M. Wang***, B. Freeman, L. Katz, 2026. Incorporating ion pairing into an extended Pitzer model for activity coefficients: implications for ion exchange membrane modeling. *International Congress on Membranes and Membrane Processes (ICOM)*, San Antonio, TX.

4. **M. Wang***, J. Cui, N. Vinh, Z. Huang, B. Freeman, L. Katz, 2025. Comprehensive analysis of ion pairing and solvation in monovalent sulfate salts: insights from Raman and dielectric spectroscopy. *American Chemical Society Spring 2025 Meeting*, San Diego, CA.

3**. M. Wang***. Mineral Scaling on Reverse Osmosis Membranes: Role of Mass, Orientation, and Crystallinity on Permeability. Invited seminar, Environmental and Water Resources Engineering (EWRE), The University of Texas at Austin, Austin, TX, February 15, 2024.

2. **M. Wang***, H. Nguyen, S. Lounder, A. Asatekin, D. Rodrigues, 2021. Calcium sulfate formation on different zwitterionic amphiphilic copolymer substrates: implications for developing anti-scaling surfaces. *ACS Southwest Regional Meeting*, Austin, TX.

1. **M. Wang***, B. Cao, Y. Hu, 2019. Effects of organic macromolecules on gypsum scaling of reverse osmosis membranes. *Water Quality Technology Conference*, Dallas, TX.

Poster Presentations (* indicates the presenter):

3. L. Katz*****, **M. Wang**, Z. Bajalan, Z. Huang, J. Cui, N. Vinh, B. Freeman, G. Henkelman, 2026. The role of ion pairing in membrane separations. *Gordon Research Conference on Environmental Sciences: Water,* Holderness, NH.

2. **M. Wang***, Z. Huang, J. Cui, N. Vinh, B. Freeman, L. Katz, 2025. Quantifying ion pairing in monovalent sulfate salts: implications for ion exchange membrane modeling. *AEESP 2025 Research and Education Conference*, Durham, NC.

1. Z. Huang*****, Z. Bajalan*****, **M. Wang**, R. Sujanani, G. Henkelman, L. Katz, B. Freeman, 2025. Effect of Ion Pairing on Salt Transport in Ion Exchange Membranes (IEMs). *The 2025 EFRC-Hub-CMS-CCS Principal Investigators' Meeting*, Rockville, MD.

ACADEMIC SERVICE

- **Journal Reviewer (105 peer reviews completed for 10+ academic journals)**
  - *npj Clean Water*
  - *Water Resources Research*
  - *Physics of Fluids*
  - *Energy Science & Engineering*
  - *Sustainability*

  - *Processes*
  - *Materials*
  - *Water*
  - *Energies*
  - *Fluids*
  - *Frontiers in Earth Science*
- **Editorial Service**
  - Guest Editor, ***Water***, Special Issue on Membrane Fouling Control Strategies in Water Treatment *2024*
- **Profession Society Membership**
  - *American Chemical Society*
  - *The Association of Environmental Engineering and Science Professors (AEESP)*
  - *The North American Membrane Society (NAMS)*

PROFESSIONAL EXPERIENCE

Consultant Intern, Baker Tilly Beijing *Beijing, China | 05/2018–07/2018*

- Supported internal audits, client proposal development, project analysis, and technical report preparation

Research Intern, ConocoPhillips China *Beijing, China | 07/2017–09/2017*

- Processed and analyzed oil-well production, drilling, and core-sample data, including data management, visualization, and sand-size distribution analysis
- Evaluated wellbore designs and proppant size selection using drilling and formation sand-size data for the Penglai Oilfield

TECHNICAL SKILLS

- **Membrane & Surface Characterization**:

  Atomic force microscopy (AFM), scanning electron microscopy (SEM), X-ray photoelectron spectroscopy (XPS), X-ray diffraction (XRD), attenuated total reflectance Fourier-transform infrared spectroscopy (ATR-FTIR), quartz crystal microbalance with dissipation (QCM-D), zeta potential analysis
- **Synchrotron & Scattering Techniques**:

  Grazing-incidence small-angle X-ray scattering (GISAXS/SAXS), grazing-incidence wide-angle X-ray scattering (GIWAXS), X-ray adsorption spectroscopy
- **Spectroscopy & Chemical Analysis**:

  Raman spectroscopy, dielectric spectroscopy, inductively coupled plasma mass spectrometry (ICP–MS and ICP–OES), atomic absorption spectroscopy (AAS), ion chromatography, total organic carbon (TOC) analysis, UV–Vis spectroscopy
- **Computational & Thermodynamic Modeling**:

  Geochemical modeling (Geochemist's Workbench), computational fluid dynamics (ANSYS Fluent, ICEM CFD, CFD-Post), thermodynamic modeling (MINTEQ)
- **Programming & Data Science**:

  Python, MATLAB, C/C++, Linux Shell, Git
- **Engineering & Design Software**:

  AutoCAD, SolidWorks

REFERENCES

Lynn E. Katz
Professor
Hussein M. Alharthy Centennial Chair in Civil Engineering
Director, Center for Water and the Environment
Associate Director, Center for Materials for Water and Energy Systems
Department of Civil, Architectural and Environmental Engineering
The University of Texas at Austin
Email: lynnkatz@mail.utexas.edu
Phone: (512) 633-4371

Debora F. Rodrigues
Professor
Dean's Professor and Chair
Department of Environmental Engineering and Earth Sciences
Clemson University
Email: dfrodri@clemson.edu
Phone: (864) 656-3519

Benny D. Freeman
Professor
William J. (Bill) Murray, Jr. Endowed Chair in Engineering
Director, Center for Materials for Water and Energy Systems
McKetta Department of Chemical Engineering
University of Texas at Austin
Email: freeman@che.utexas.edu
Phone: (512) 626-6612

Ayse Asatekin
Professor
Chemical and Biological Engineering
Tufts University
Email: ayse.asatekin@tufts.edu
Phone: (617) 627-4681

Vinh Q. Nguyen
Associate Professor
Department of Physics
Virginia Tech
Email: vinh@vt.edu
Phone: (540) 231-3158